

Charan Vardham

216-804-5690 | charanvardham@gmail.com | github.com/Charan6924

SUMMARY

Applied researcher and CS undergraduate with hands-on experience building and training large-scale LLM and deep learning systems, designing RAG pipelines, and deploying AI-powered developer tools. Passionate about advancing AI for software engineering through rigorous ML experimentation, dataset construction, and post-training model research. Experienced collaborating across cross-functional teams to take research projects from prototype to production.

EDUCATION

University of Wisconsin Madison

Madison, WI

Bachelor of Science in Computer Science, Minor in Mathematics

Aug. 2024 – May 2028

- **Relevant Courses:** Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Computer Security

EXPERIENCE

AI/ML Intern

May 2025 – Aug. 2025

IQVIA

Remote

- Designed and deployed a production RAG pipeline over 200MB of clinical documents, integrating document ingestion, text chunking, embedding generation, and retrieval stages to achieve 90% retrieval accuracy
- Built LLM-backed medical chatbot with MongoDB-backed chat history, minimizing hallucinations and improving factual grounding for downstream developer and clinical users
- Integrated NVIDIA Riva APIs for real-time STT/TTS in a voice-enabled LLM assistant, demonstrating end-to-end AI for developer tooling across modalities

Undergraduate Applied Researcher

Aug. 2025 – Present

Case Western Reserve University

Cleveland, OH

- Investigated deep learning kernel transfer methods for CT imaging using CNN models trained on 5000+ scan slices in PyTorch, evaluating reconstruction quality via PSNR and SSIM metrics
- Managed large-scale ML experiments end-to-end including dataset construction, model training, and systematic evaluation; preparing results for a research publication

xLab Applied Research Intern

Jan. 2026 – Present

Weatherhead School of Management, CWRU

Cleveland, OH

- Developing an AI-powered platform using LLMs and Anthropic/OpenAI APIs to simulate interactive business decision-making scenarios, taking the project from research concept to deployed product
- Built full-stack application with Next.js (TypeScript), Tailwind CSS, PostgreSQL, and Prisma, integrating agentic LLM workflows with dynamic HeyGen avatar interfaces

PROJECTS

GPT-2 Pretraining from Scratch | *PyTorch, Python, CUDA, SLURM, H100 GPUs*

Feb. 2026

- Pretrained GPT-2 (124M parameters) on 10B tokens using NVIDIA H100 GPUs, achieving final validation loss of 3.0 at 260K tokens/sec throughput via bfloat16 mixed-precision and batch scheduling
- Gained deep familiarity with large-scale model training infrastructure, post-training optimization, and code-specific LLM architectures directly relevant to code completion and code embedding research

Real-Time Fraud Detection ML System | *Python, XGBoost, AWS Lambda, Docker, Redis*

June 2026

- Built end-to-end ML pipeline with sub-100ms inference using a containerized XGBoost model on AWS Lambda, with a TypeScript REST inference API and an online/offline feature store using Redis and S3/Parquet
- Implemented CI/CD pipeline on GitHub Actions with automated retraining on data drift, canary rollouts via weighted Lambda aliases, and promotion gates, demonstrating production-grade ML experimentation practices

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R, HTML/CSS

AI/ML Frameworks: PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers, Scikit-Learn, XGBoost, Pandas, NumPy

LLM and AI Tools: OpenAI API, Anthropic API, RAG pipelines, LLM fine-tuning, prompt engineering, code embedding models, agentic workflows, NVIDIA Riva

Engineering and Deployment: Git, Docker, AWS, Kubernetes, CI/CD, GitHub Actions, REST APIs, Next.js, FastAPI, PostgreSQL, Redis, SLURM